

Github link: <https://github.com/chenjie-gu/ds5500>

Weekly Update — NSF Funding Analysis

1. What did you accomplish this week?

I extended the topic modeling so it runs separately for each division, OAC, CNS, CCF, and IIS, for both LDA and BERTopic. Now each division gets its own set of topics instead of being squeezed into one shared topic space. For LDA, each division runs its own coherence sweep and picks its own best k, which ranged from 4 for OAC up to 20 for CNS. For BERTopic I reused the embeddings I already computed and just sliced them by division instead of re-encoding, which kept the runtime down.

I also reworked the subfield charts to drop the year axis and show aggregated totals instead, since five years is too short to read a real trend from. I cut the institution and PI charts down to top 100 schools, top 50 bubbles, and top 30 PI pairs, and added the share of total funding and awards each group covers right in the chart subtitle. Finally I rebuilt the state maps so Alaska and Hawaii sit in shrunken insets and the lower 48 fills the frame, and I added the actual campus locations as points sized by funding.

2. Did you learn or discover anything new?

Splitting the topics by division showed structure the pooled model was hiding. CNS clearly breaks into separate wireless, security, and smart infrastructure groups, and CCF separates theory, hardware, and quantum work cleanly. The most useful new view was comparing funding against award count side by side. Conference, workshop, and outreach topics rank high on number of awards but low on dollars, which lines up with them being lots of small grants, while infrastructure and simulation topics are the reverse. The institution numbers showed the concentration is real but not extreme. The top 100 schools hold about 77 percent of funding and 72 percent of awards, and the top 50 schools in the bubble chart cover about 57 percent of funding. The overall decline I found last week still holds across every way I sliced the data.

3. Are you uncertain, stuck, or worried about anything?

The per division BERTopic models are probably too granular. The smaller divisions produce more than 25 topics each, and a lot of them only have 15 to 25 awards with one word labels that are hard to read, so those charts need a higher cluster size threshold or some manual relabeling before they are presentable. The manual relabeling problem from last week still applies to all the LDA topics too, and a couple of the per division LDA topics came out with low coherence and messy terms in the label. I am also still not sure the 2025 data is complete, which would make the decline look steeper than it really is. The campus geocoding works well for the big schools, but smaller institutions fall back to the center of their state instead of a real campus location.

4. What is your plan for next week?

Clean up or combine the per division BERTopic labels, finish the markdown write up for each section, and finalize the PDF. The core analysis is done, so the rest is mostly presentation and interpretation.

5. Anything else?

No, on track.